

A*

Log

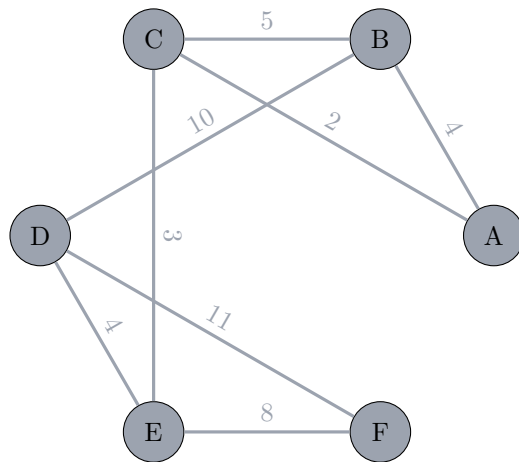
Initial state

A*

Log

Initial state

Graph

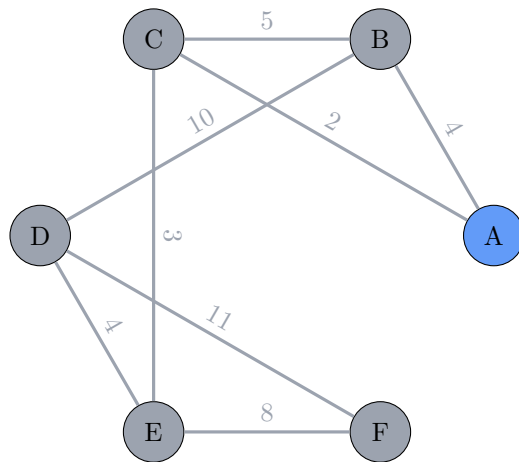


A*

Log

Initial state

Graph

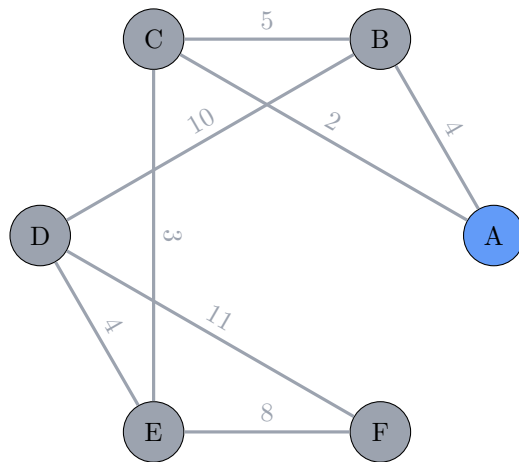


A*

Log

Initial state

Graph



Open Set

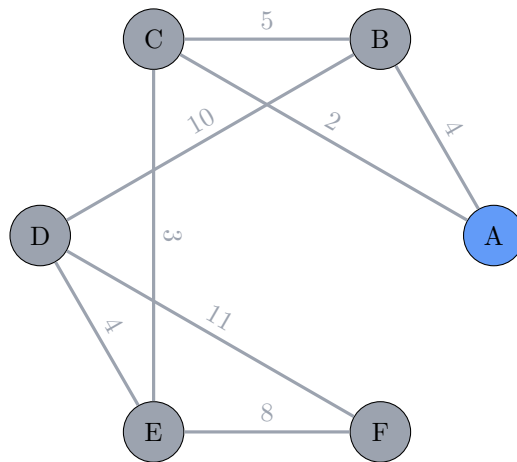
A

A*

Log

Initial state

Graph



Open Set

A

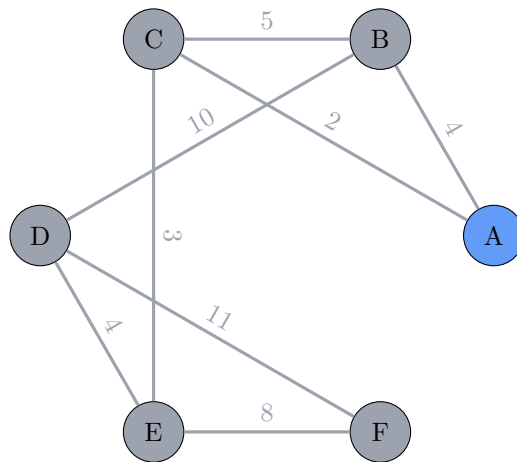
Closed Set

A*

Log

Initial state

Graph



Open Set

A

Closed Set

Node Costs

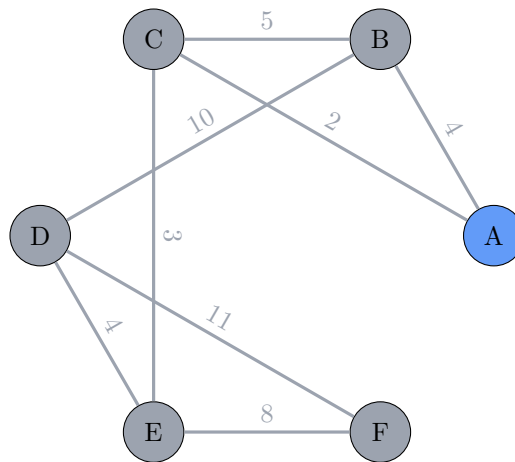
0	0	0	0	0	0
A	B	C	D	E	F

A*

Log

Selecting the node with the lowest estimated total cost $f(n) = g(n) + h(n)$.

Graph



Open Set

A

Closed Set

Node Costs

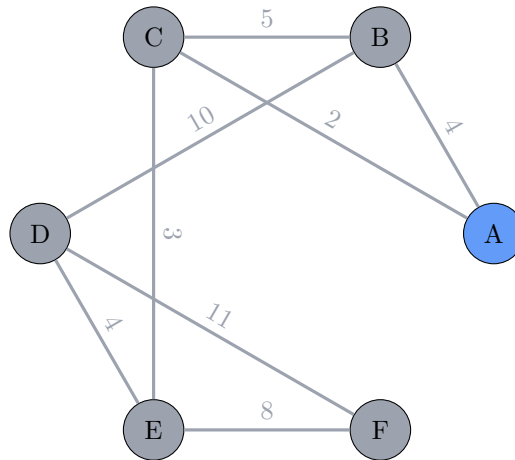
0	0	0	0	0	0
A	B	C	D	E	F

A*

Log

Node A has the lowest estimated total cost.

Graph



Open Set

A

Closed Set

Node Costs

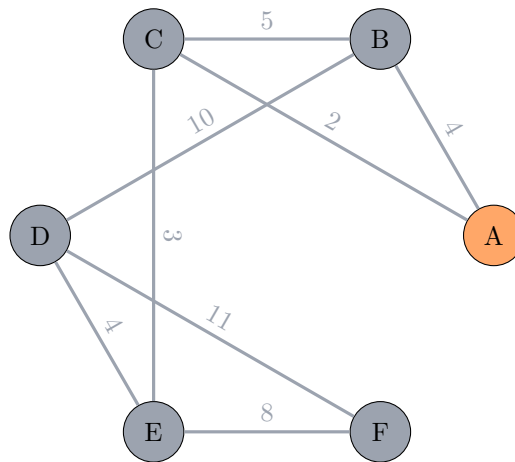
0	0	0	0	0	0
A	B	C	D	E	F

A*

Log

Exploring node A.

Graph



Open Set

Closed Set



Node Costs

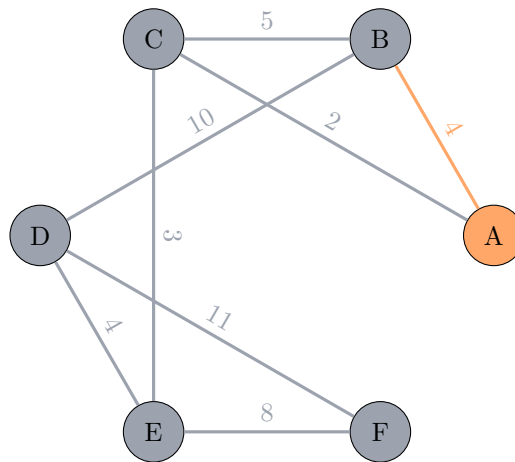
0	0	0	0	0	0
A	B	C	D	E	F

A*

Log

Checking whether reaching B through A gives a cheaper path.

Graph



Open Set

Closed Set

A

Node Costs

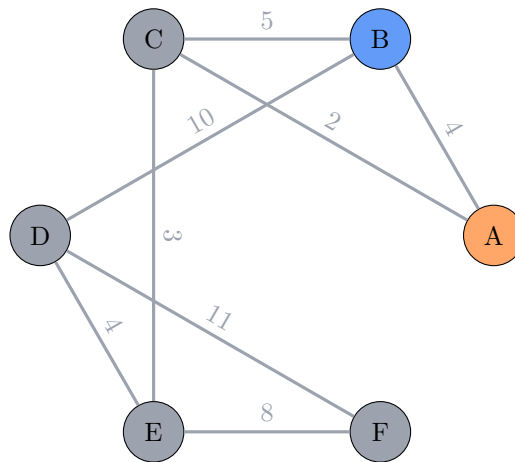
0	0	0	0	0	0
A	B	C	D	E	F

A*

Log

Node B has a promising path, so it is added to the open set.

Graph



Open Set

B

Closed Set

A

Node Costs

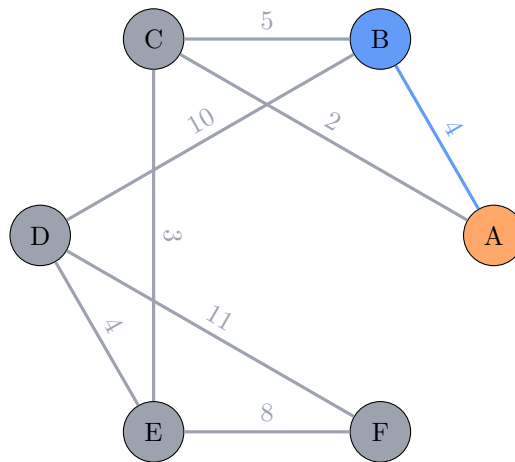
0	0	0	0	0	0
A	B	C	D	E	F

A*

Log

Found a cheaper path to B through A. Its cost is now 4.

Graph



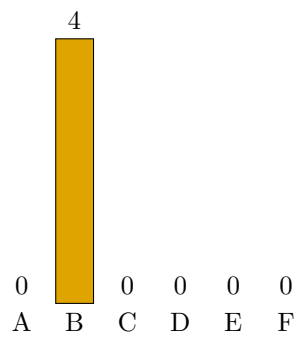
Open Set



Closed Set



Node Costs

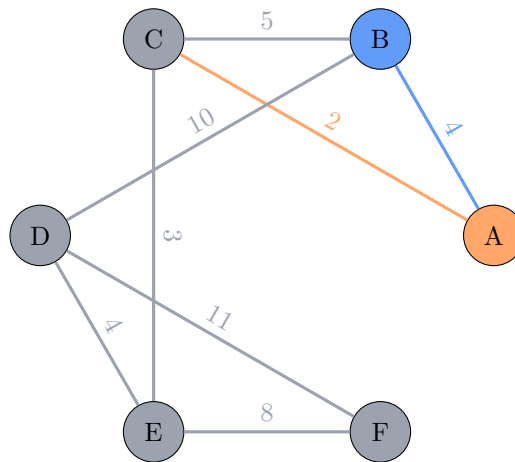


A*

Log

Checking whether reaching C through A gives a cheaper path.

Graph



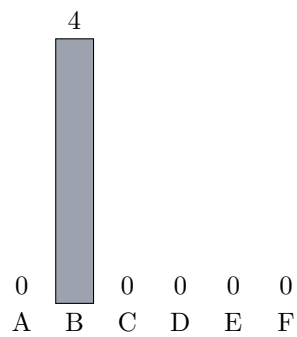
Open Set

B

Closed Set

A

Node Costs

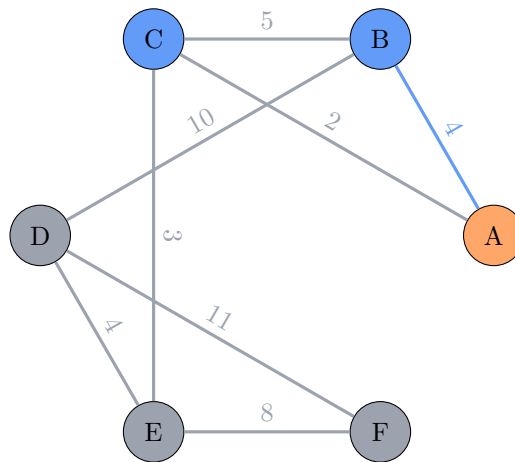


A*

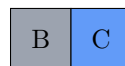
Log

Node C has a promising path, so it is added to the open set.

Graph



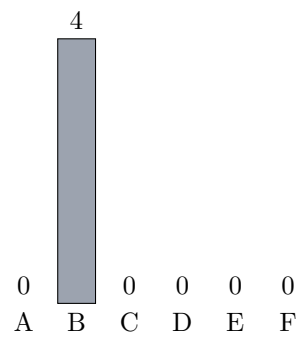
Open Set



Closed Set



Node Costs

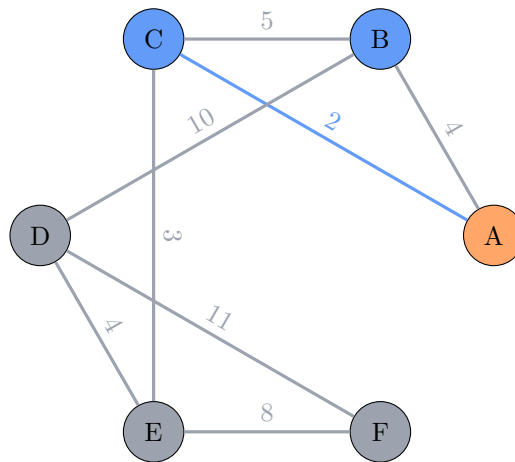


A*

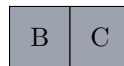
Log

Found a cheaper path to C through A. Its cost is now 2.

Graph



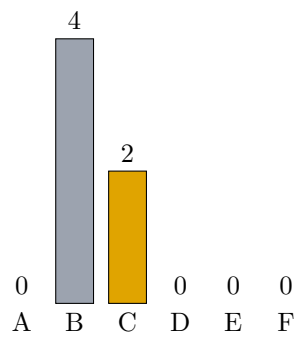
Open Set



Closed Set



Node Costs

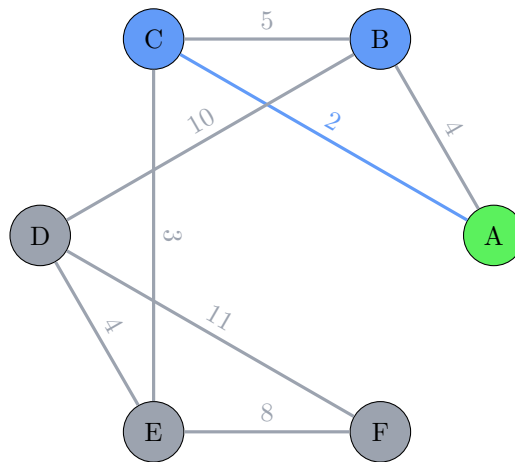


A*

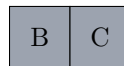
Log

Found a cheaper path to C through A. Its cost is now 2.

Graph



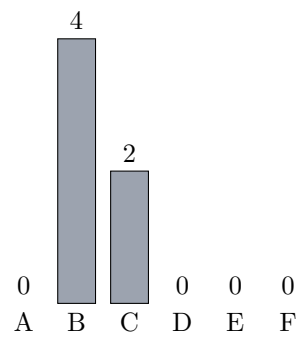
Open Set



Closed Set



Node Costs

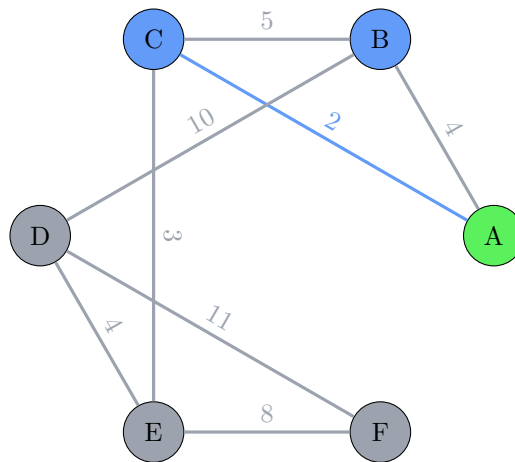


A*

Log

Selecting the node with the lowest estimated total cost $f(n) = g(n) + h(n)$.

Graph



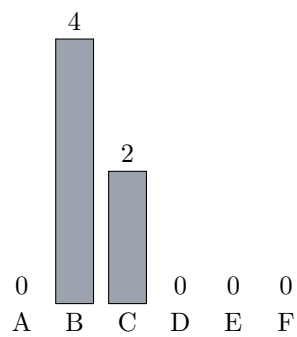
Open Set



Closed Set



Node Costs

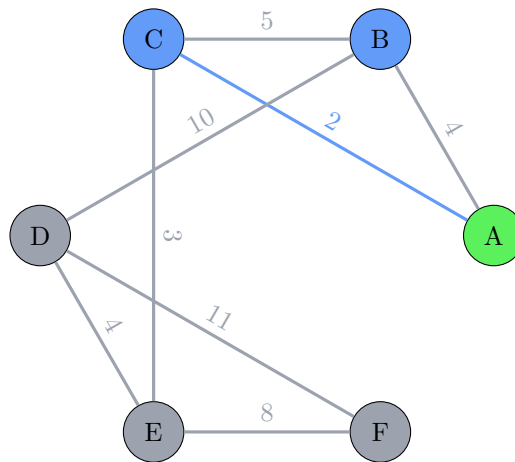


A*

Log

Node C has the lowest estimated total cost.

Graph



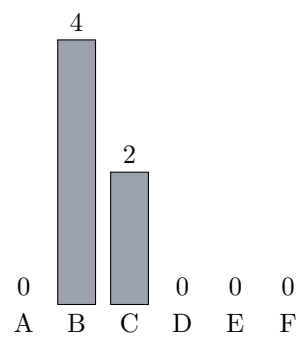
Open Set



Closed Set



Node Costs

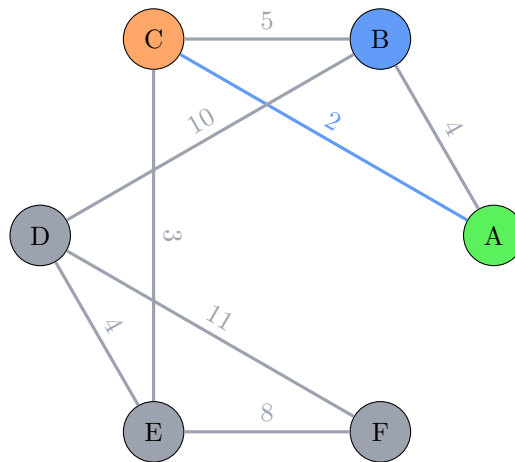


A*

Log

Exploring node C.

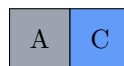
Graph



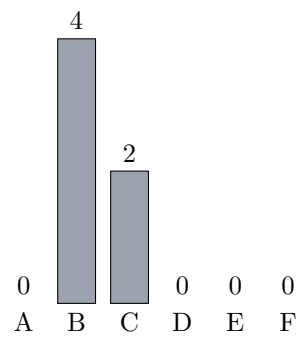
Open Set



Closed Set



Node Costs

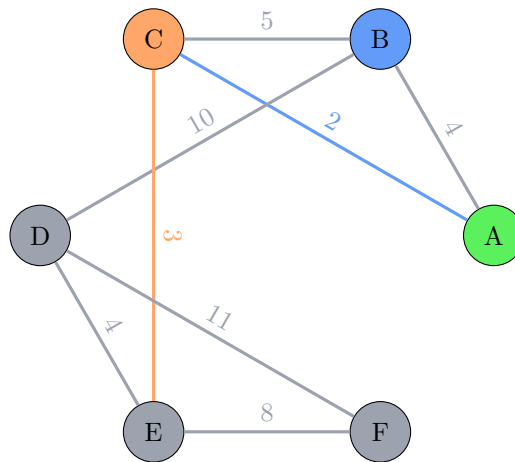


A*

Log

Checking whether reaching E through C gives a cheaper path.

Graph



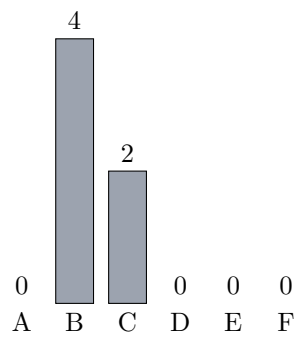
Open Set

B

Closed Set

A C

Node Costs

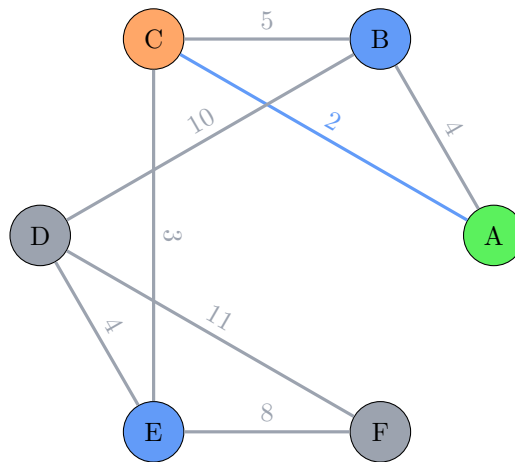


A*

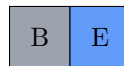
Log

Node E has a promising path, so it is added to the open set.

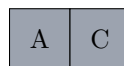
Graph



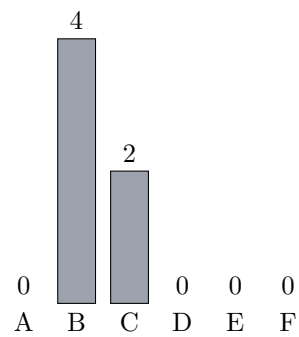
Open Set



Closed Set



Node Costs

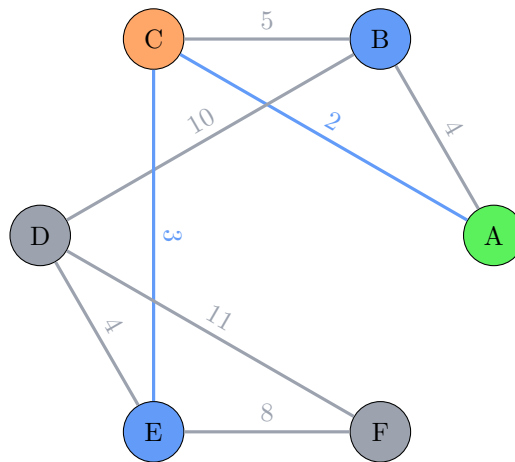


A*

Log

Found a cheaper path to E through C. Its cost is now 5.

Graph



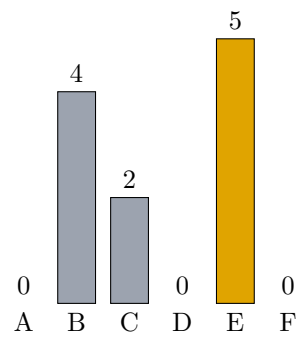
Open Set

B	E
---	---

Closed Set

A	C
---	---

Node Costs

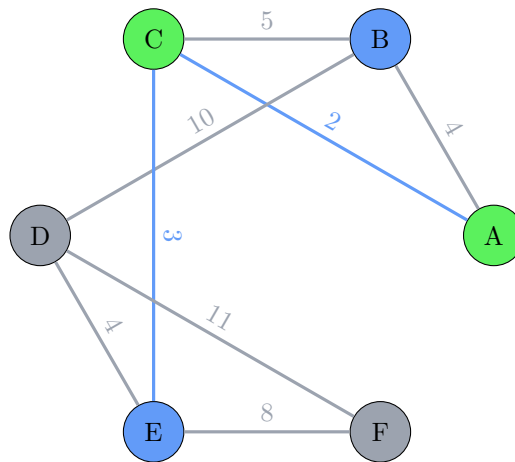


A*

Log

Found a cheaper path to E through C. Its cost is now 5.

Graph



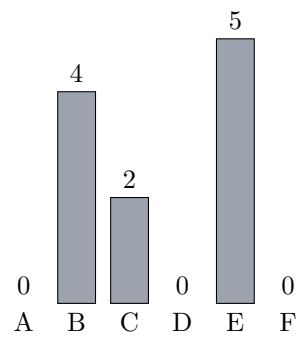
Open Set

B	E
---	---

Closed Set

A	C
---	---

Node Costs

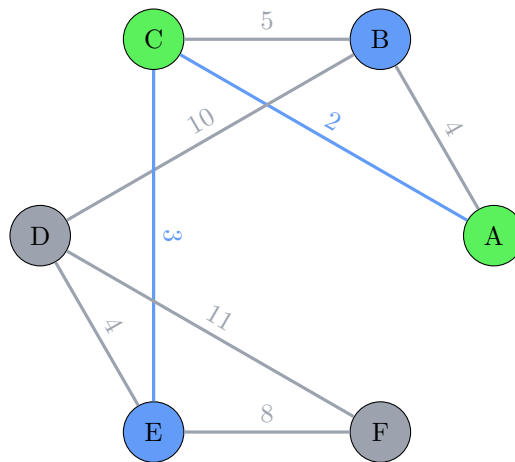


A*

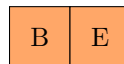
Log

Selecting the node with the lowest estimated total cost $f(n) = g(n) + h(n)$.

Graph



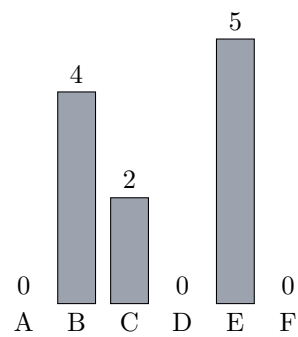
Open Set



Closed Set



Node Costs

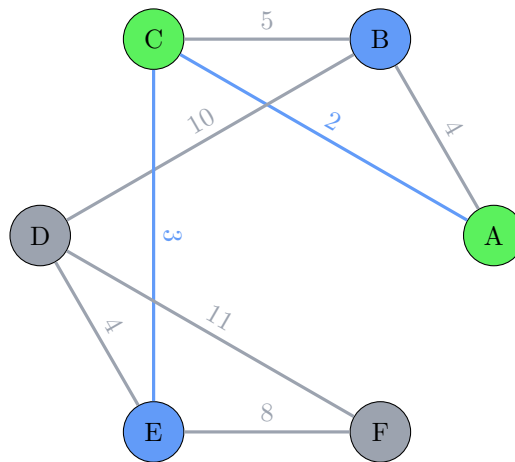


A*

Log

Node B has the lowest estimated total cost.

Graph



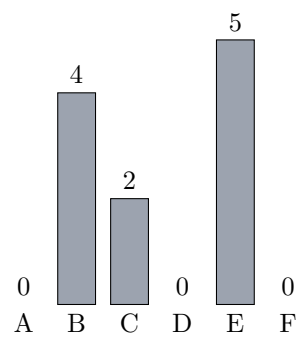
Open Set



Closed Set



Node Costs

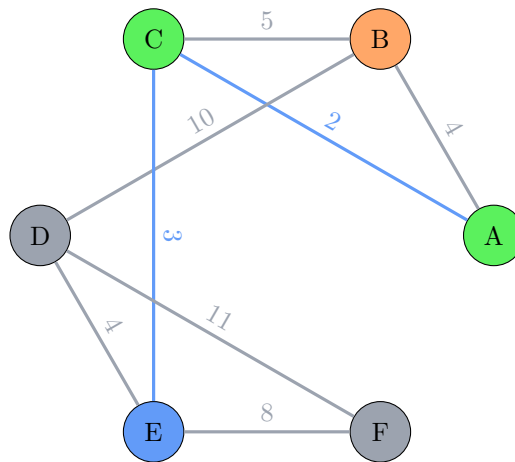


A*

Log

Exploring node B.

Graph



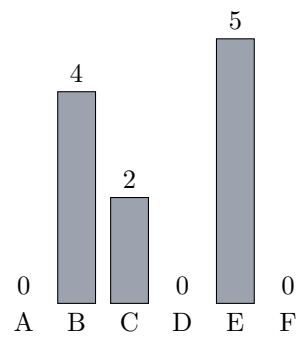
Open Set

E

Closed Set

A C B

Node Costs

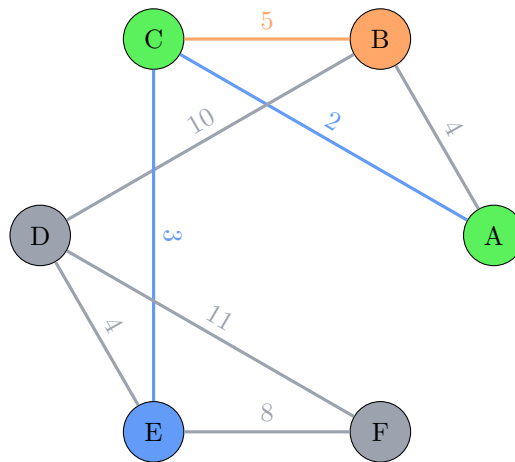


A*

Log

Checking whether reaching C through B gives a cheaper path.

Graph



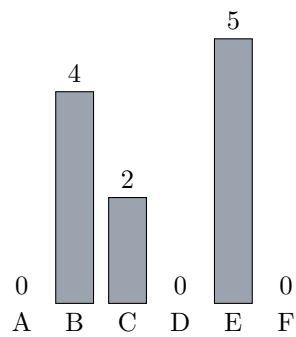
Open Set

E

Closed Set

A C B

Node Costs

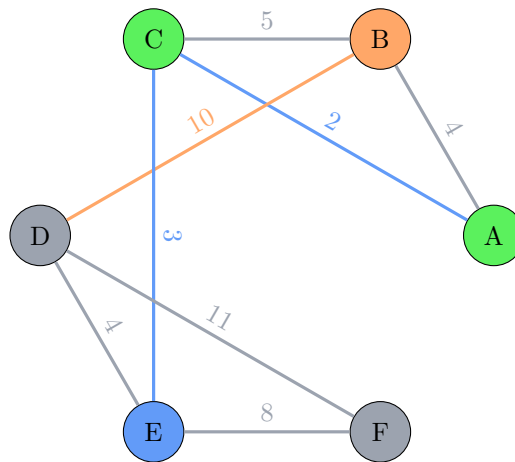


A*

Log

Checking whether reaching D through B gives a cheaper path.

Graph



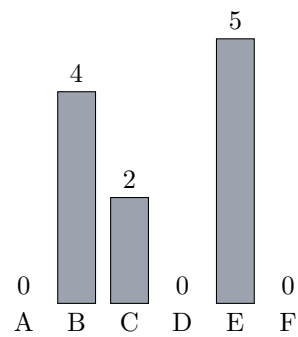
Open Set

E

Closed Set

A	C	B
---	---	---

Node Costs

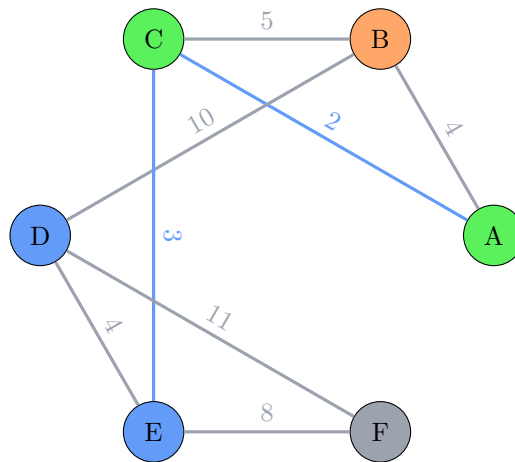


A*

Log

Node D has a promising path, so it is added to the open set.

Graph



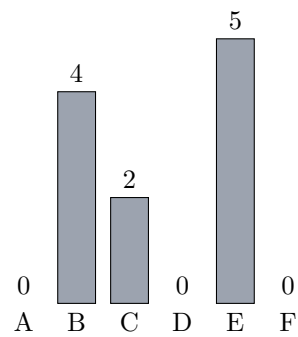
Open Set



Closed Set



Node Costs

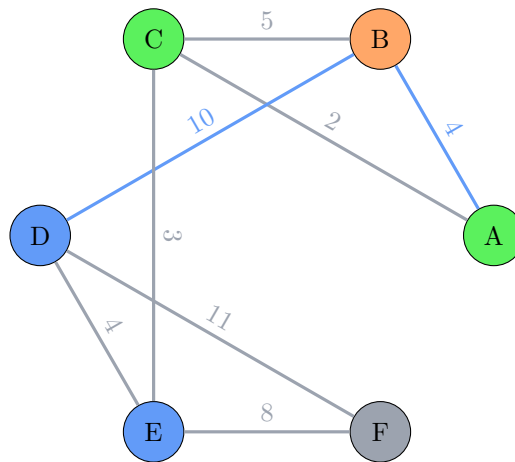


A*

Log

Found a cheaper path to D through B. Its cost is now 14.

Graph



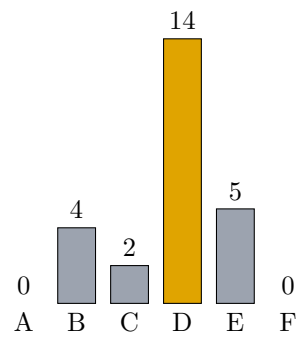
Open Set

E	D
---	---

Closed Set

A	C	B
---	---	---

Node Costs

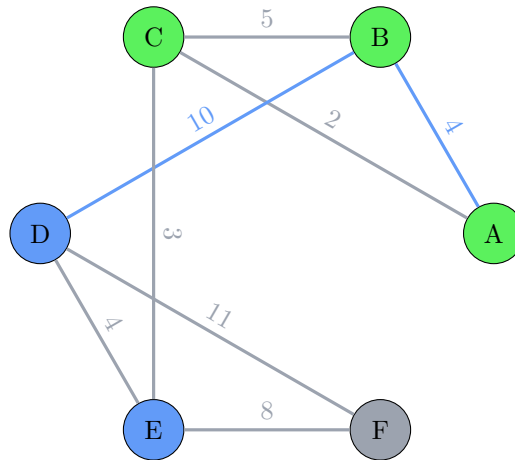


A*

Log

Found a cheaper path to D through B. Its cost is now 14.

Graph



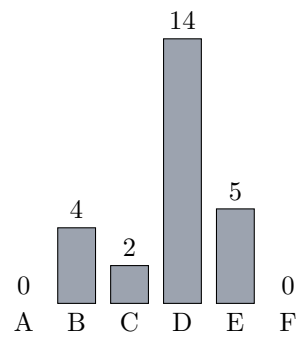
Open Set

E	D
---	---

Closed Set

A	C	B
---	---	---

Node Costs

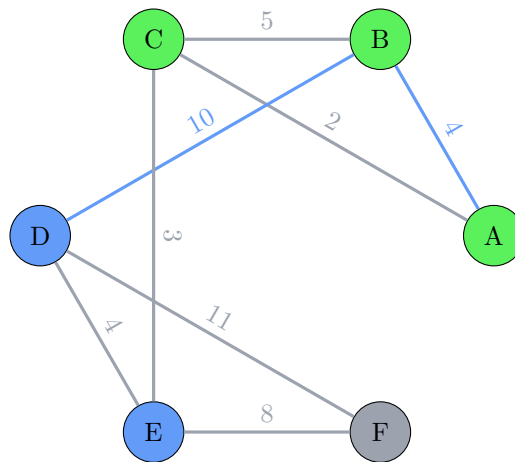


A*

Log

Selecting the node with the lowest estimated total cost $f(n) = g(n) + h(n)$.

Graph



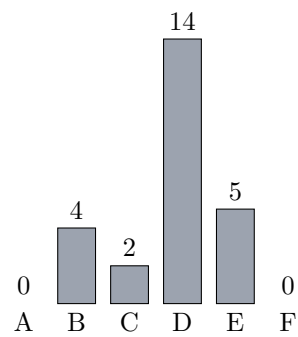
Open Set



Closed Set



Node Costs

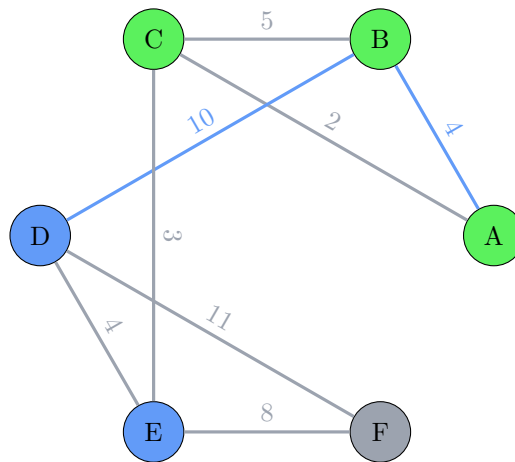


A*

Log

Node E has the lowest estimated total cost.

Graph



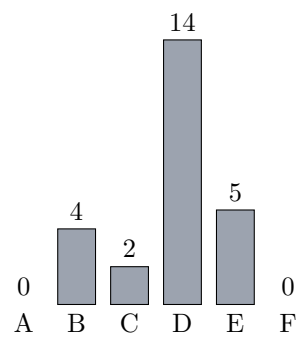
Open Set



Closed Set



Node Costs

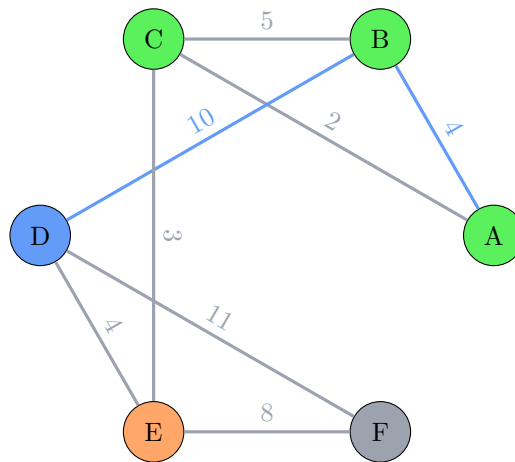


A*

Log

Exploring node E.

Graph



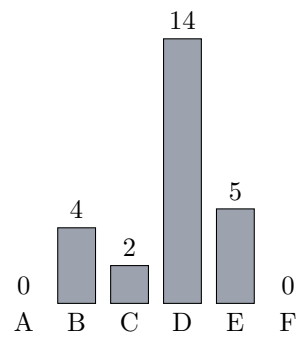
Open Set

D

Closed Set

A	C	B	E
---	---	---	---

Node Costs

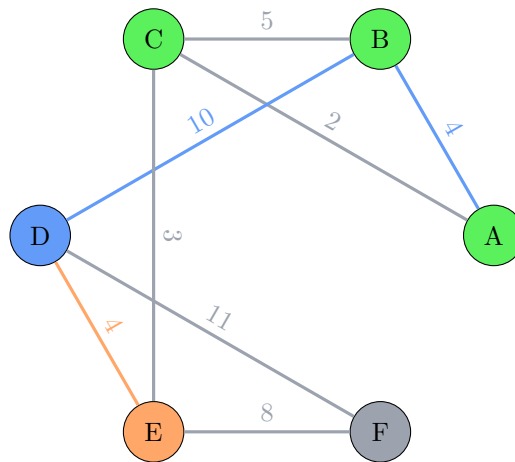


A*

Log

Checking whether reaching D through E gives a cheaper path.

Graph



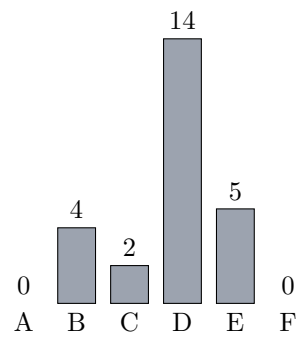
Open Set

D

Closed Set

A	C	B	E
---	---	---	---

Node Costs

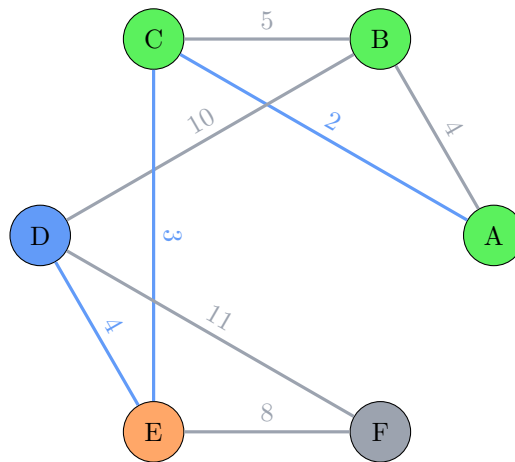


A*

Log

Found a cheaper path to D through E. Its cost is now 9.

Graph



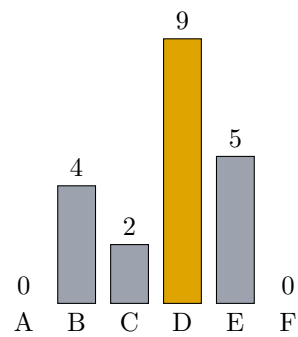
Open Set

D

Closed Set

A	C	B	E
---	---	---	---

Node Costs

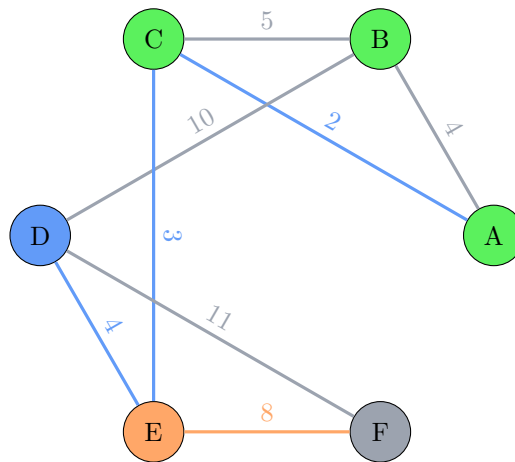


A*

Log

Checking whether reaching F through E gives a cheaper path.

Graph



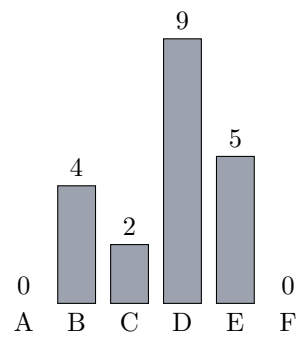
Open Set

D

Closed Set

A	C	B	E
---	---	---	---

Node Costs

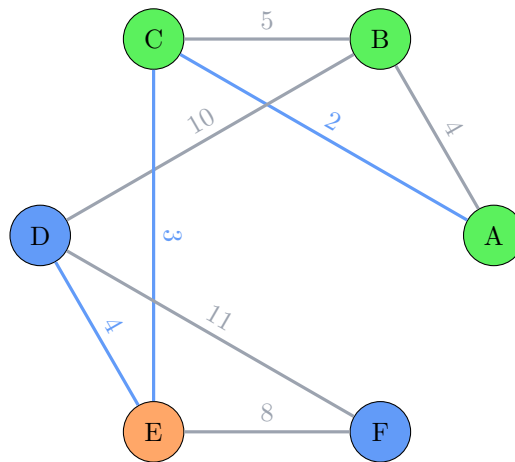


A*

Log

Node F has a promising path, so it is added to the open set.

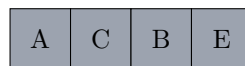
Graph



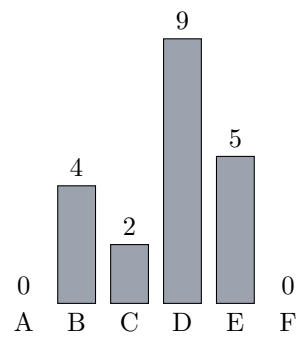
Open Set



Closed Set



Node Costs

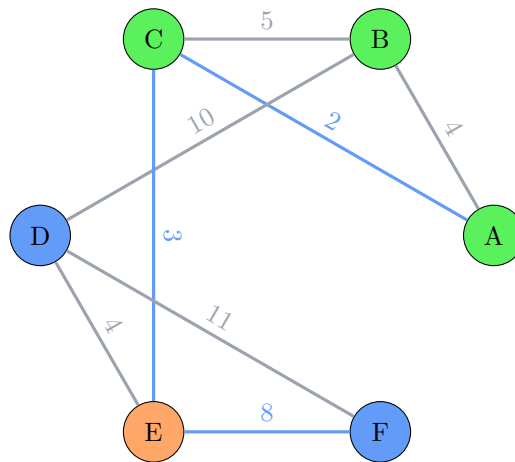


A*

Log

Found a cheaper path to F through E. Its cost is now 13.

Graph



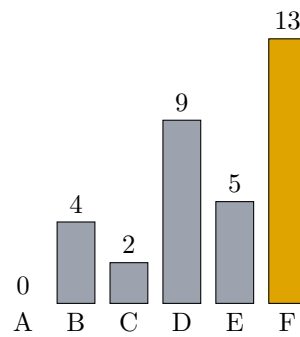
Open Set

D	F
---	---

Closed Set

A	C	B	E
---	---	---	---

Node Costs

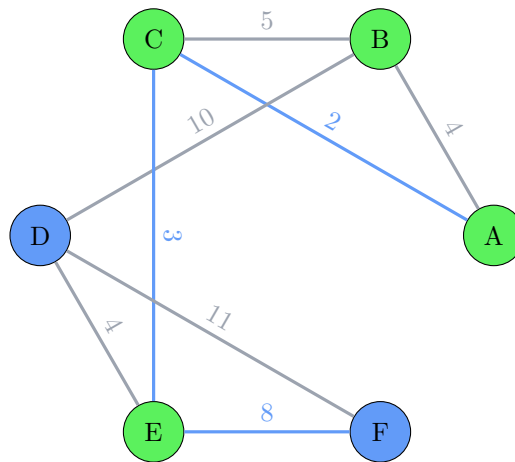


A*

Log

Found a cheaper path to F through E. Its cost is now 13.

Graph



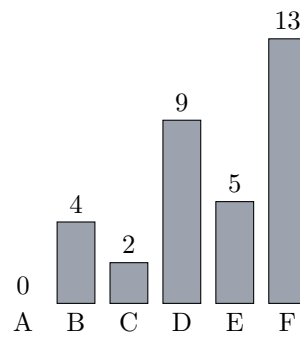
Open Set

D	F
---	---

Closed Set

A	C	B	E
---	---	---	---

Node Costs

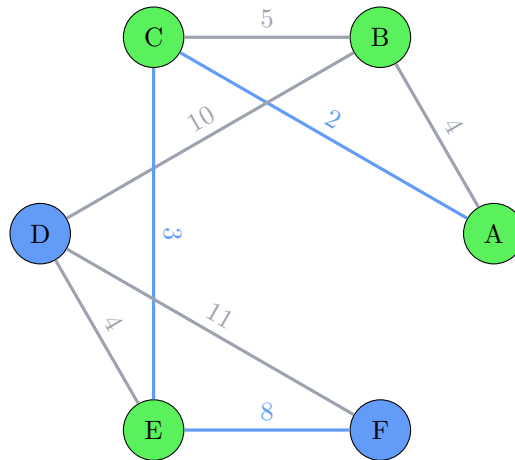


A*

Log

Selecting the node with the lowest estimated total cost $f(n) = g(n) + h(n)$.

Graph



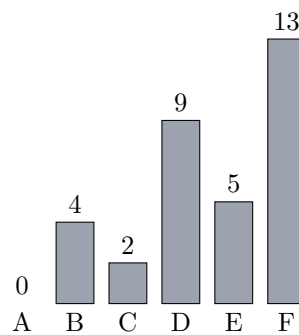
Open Set



Closed Set



Node Costs

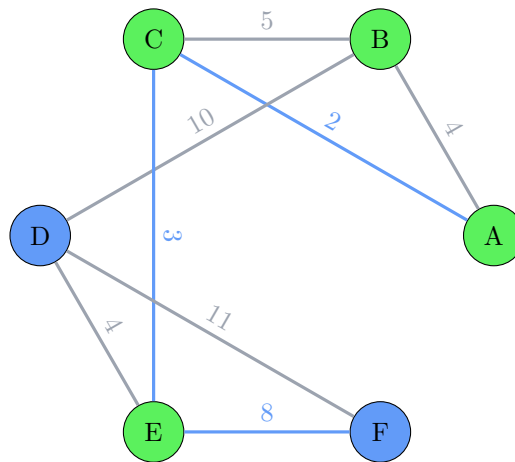


A*

Log

Node D has the lowest estimated total cost.

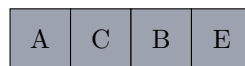
Graph



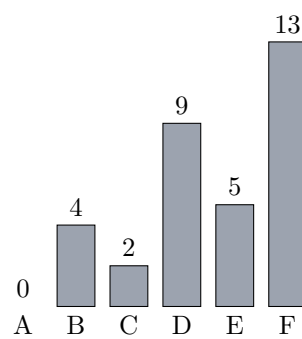
Open Set



Closed Set



Node Costs

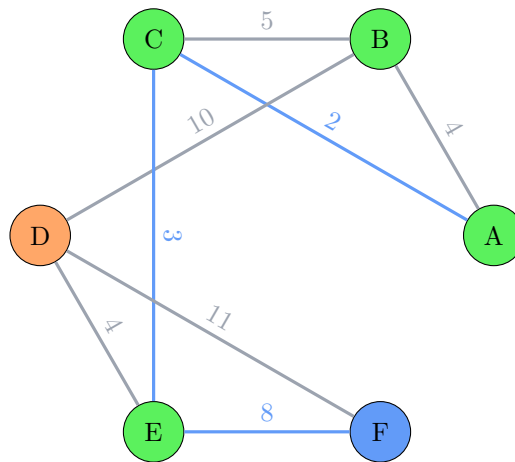


A*

Log

Exploring node D.

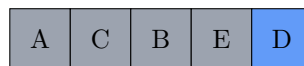
Graph



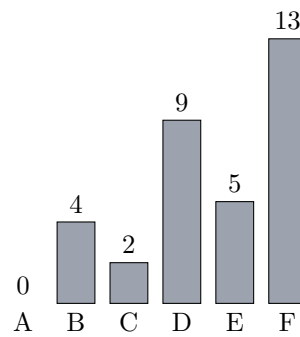
Open Set



Closed Set



Node Costs

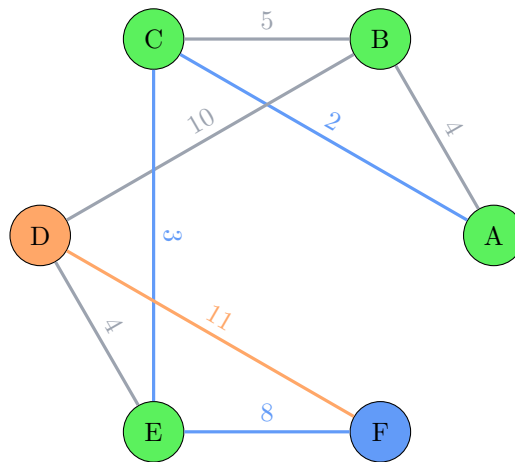


A*

Log

Checking whether reaching F through D gives a cheaper path.

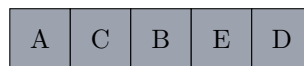
Graph



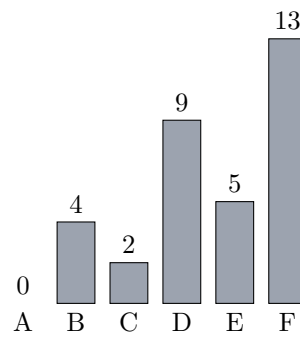
Open Set



Closed Set



Node Costs

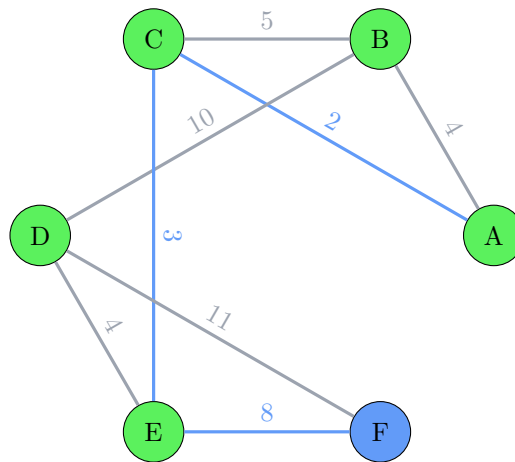


A*

Log

Checking whether reaching F through D gives a cheaper path.

Graph



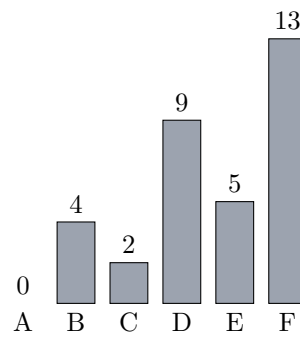
Open Set

F

Closed Set

A	C	B	E	D
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Node Costs

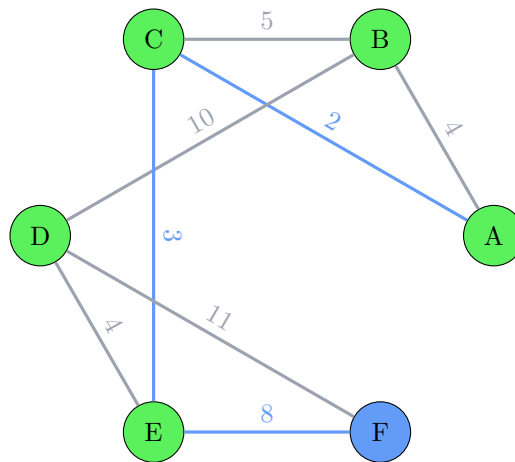


A*

Log

Selecting the node with the lowest estimated total cost $f(n) = g(n) + h(n)$.

Graph



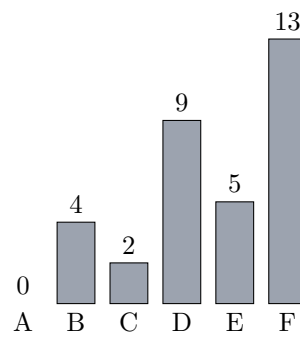
Open Set

F

Closed Set

A	C	B	E	D
---	---	---	---	---

Node Costs

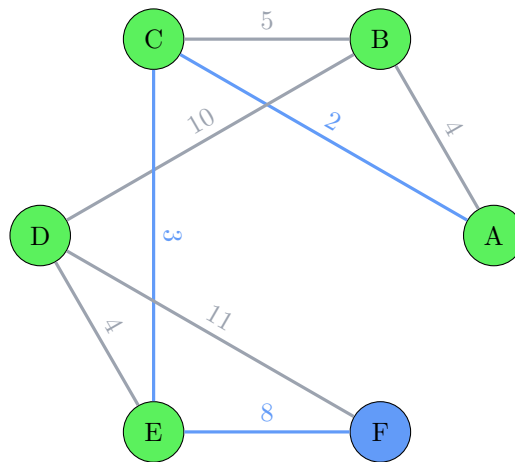


A*

Log

Node F has the lowest estimated total cost.

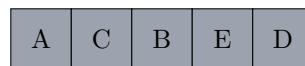
Graph



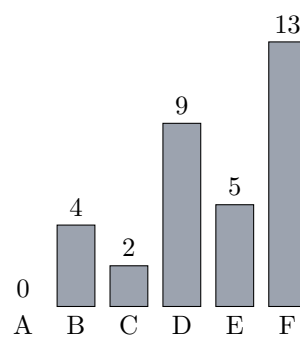
Open Set



Closed Set



Node Costs

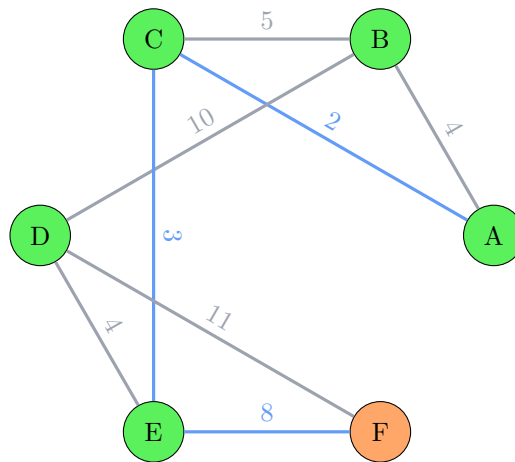


A*

Log

Exploring node F.

Graph

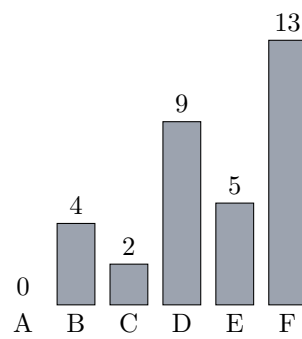


Open Set

Closed Set

A	C	B	E	D	F
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Node Costs

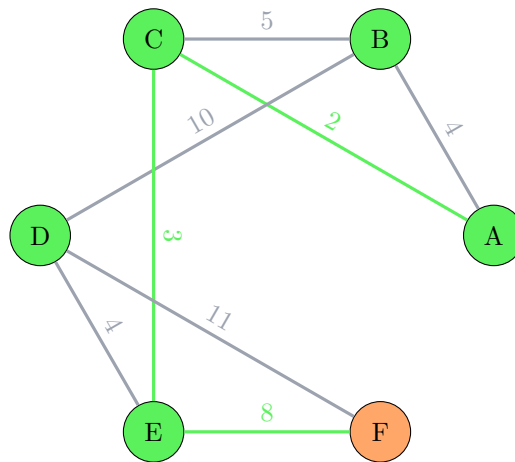


A*

Log

Reached the destination F. Tracing back the best path found.

Graph

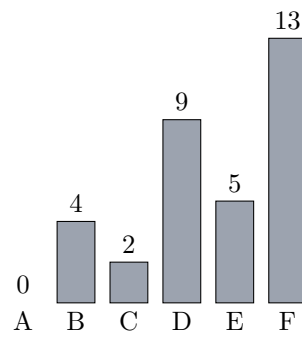


Open Set

Closed Set

A	C	B	E	D	F
---	---	---	---	---	---

Node Costs

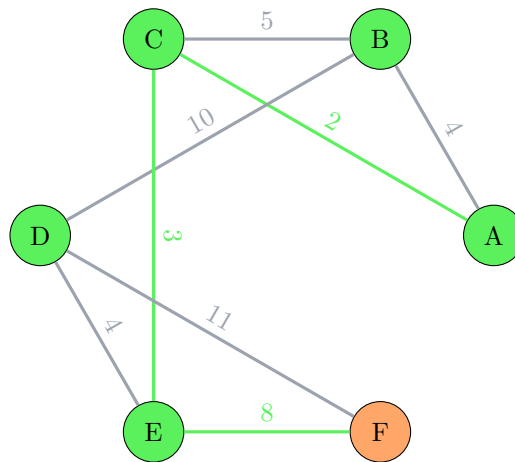


A*

Log

A* has found a path from A to F.

Graph



Open Set

Closed Set

A	C	B	E	D	F
---	---	---	---	---	---

Node Costs

